

# Preliminary Study for Autonomous Evolutionary Multitasking Method Design

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**Abstract**—Evolutionary Multi-task Optimization (EMTO) enhances optimization performance by sharing knowledge across optimization tasks. While existing knowledge transfer methods can improve optimization performance, they often require significant domain expertise that consume huge expert resources. To handle this problem, we introduce a preliminary LLM-assisted framework that autonomously designs knowledge transfer methods for EMTO without human intervention. Empirical studies on WCCI competition benchmarks demonstrate that the knowledge transfer methods designed by our proposed framework can achieve superior performance against existing hand-crafted knowledge transfer approaches.

**Index Terms**—Automatic Transfer Optimization, LLM-assisted Algorithm Design

## I. INTRODUCTION

THE paradigm of Evolutionary Multi-task Optimization (EMTO) shows advances by facilitating the knowledge sharing across multiple related optimization tasks [1]. Central to EMTO's success are the knowledge transfer methods that capture knowledge building-blocks for improved search performance. However, existing EMTO's designs often demand deep domain expertise and extensive manual efforts [1].

Studies on large language models (LLMs) have revealed their strong aptitude for optimization tasks, encompassing both combinatorial and numerical domains [2]. However, while these applications show good results, they often suffer from scalability issues when applied to high-dimensional optimization problems [3]. To overcome these limitations, researchers have begun using LLMs not as solvers, but as autonomous agents that design or enhance optimization strategies [4]. Inspired by this direction, we propose an LLM-assisted framework that autonomously generates effective and efficient knowledge transfer models for EMTO tasks. The framework employs few-shot chain-of-thought prompts to guide LLMs through the design space, enabling automatic exploration of novel structures within KTM. Furthermore, the

proposed framework also integrates a multi-objective optimization method that balances transfer quality with computational cost, ensuring both effectiveness and efficiency are considered. Our contributions include:

- Introducing the LLM-driven framework for autonomous KTM design in EMTO.
- Applying prompt engineering to generate scalable and high-performing KTMs without expert knowledge.
- Empirical studies demonstrate that the generated models outperform the hand-crafted approaches across diverse EMTO scenarios.

## II. METHODOLOGY

The paper proposes an optimization approach using large language models (LLMs) to facilitate the autonomous generation of high-quality knowledge transfer methods (KTMs) for EMTO. By formulating KTM generation as a multi-objective search problem, the framework simultaneously optimizes for effectiveness (task performance) and efficiency (computational runtime), offering a scalable alternative to hand-crafted transfer methods. At the heart of the framework is a general-purpose LLM prompted to generate executable KTM designs. To compensate for the LLM's lack of inherent knowledge about EMTO, we develop a structured prompting mechanism called Few-Shot Chain-of-Thought (FSCOT), which guides the LLM through a reasoned sequence of design steps, encouraging the generation of clear, adaptable, and efficient algorithm logic. FSCOT prompts define elements such as problem pairing, transfer strategies, and solution transformation within defined constraints—ensuring functional and interpretable output.

The proposed framework begins with the LLM generating an initial population of candidate KTMs using FSCOT-informed prompts. Each KTM is evaluated across a suite of EMTO tasks, with its normalized value and runtime recorded.

A non-dominated sorting strategy is applied to identify Pareto-optimal KTM candidates that balance both objectives. In each generation, effective KTMs are selected via roulette wheel selection. Next, the LLM will be prompted to synthesize new KTMs based on the structure and performance of these selected parents. With a small probability, mutation is introduced by prompting the LLM to revise an existing powerful KTM, maintaining the format while introducing exploratory changes. In the KTM evolution loop, generated KTMs are continuously evaluated and compared, with the poorest KTMs in the population replaced to preserve competitive selection pressure. This evolution loop proceeds for a predefined number of generations, resulting in a final set of KTMs, which reflect the optimal trade-off between search performance and computational cost.

Through the integration of multi-objective optimization, guided prompt engineering, and iterative KTM evolution, the proposed framework enables autonomous discovery of efficient and effective transfer strategies—eliminating manual effort across diverse EMTO scenarios.

### III. EXPERIMENTAL STUDY

TABLE I: Comparison of normalized fitness values and average execution times for VCM, SMM, and the top-performing KTM (KTM\*), automatically generated by the proposed method, tested on MTSSO benchmarks across 10 runs. The superior result in each case is indicated in bold.

| Benchmark | VCM   |              | SMM   |        | KTM*        |              |
|-----------|-------|--------------|-------|--------|-------------|--------------|
|           | Nor.V | Time         | Nor.V | Time   | Nor.V       | Time         |
| WCCI1     | 0.81  | 76.53        | 0.55  | 94.84  | <b>0.19</b> | <b>67.79</b> |
| WCCI2     | 0.75  | 81.82        | 0.20  | 103.41 | <b>0.03</b> | <b>77.25</b> |
| WCCI3     | 0.96  | <b>79.64</b> | 0.88  | 99.25  | <b>0.69</b> | 94.85        |
| WCCI4     | 0.89  | 80.01        | 0.83  | 100.00 | <b>0.45</b> | <b>69.30</b> |
| WCCI5     | 0.98  | <b>83.19</b> | 0.92  | 103.79 | <b>0.78</b> | 126.65       |
| WCCI6     | 1.00  | 80.26        | 0.86  | 103.01 | <b>0.69</b> | <b>69.16</b> |
| WCCI7     | 0.89  | <b>84.93</b> | 0.84  | 106.24 | <b>0.70</b> | 242.21       |
| WCCI8     | 0.91  | <b>84.34</b> | 0.94  | 104.39 | <b>0.80</b> | 87.40        |
| WCCI9     | 0.93  | 83.32        | 0.96  | 102.71 | <b>0.71</b> | <b>74.55</b> |
| WCCI10    | 0.94  | 83.74        | 0.98  | 103.22 | <b>0.82</b> | <b>75.35</b> |

The proposed framework was evaluated on the CEC2024 MTSSO test suite, which includes ten benchmarks of 50 diverse 50-D tasks derived from classic functions (e.g., Sphere, Rosenbrock, Ackley). The proposed KTM is compared with two hand-crafted methods: Vertical Crossover (VCM) [5] and Solution Mapping with Autoencoder (SMM) [6]. All methods use the same optimizer [7] with a population of 100, 100 generations, and 10,000 evaluation times per task. Our framework maintains a KTM population of 10 for up to 10 generations, and the LLM runs at temperature 0.5 for prompt-driven code generation. Performance is measured by normalized values and runtime (lower is better).

Table I summarizes the performance of the most effective KTM (i.e., KTM\*) across WCCI benchmarks, based on the lowest normalized values over 10 runs, compared to VCM and SMM. While VCM had the shortest runtimes but weakest transfer ability, SMM offered slight improvements with longer processing. KTM\* consistently achieved superior results in both ‘Nor.V’ and ‘Time’. For instance, on WCCI1, it yielded significantly better metrics than both baselines. Competitive results can also be observed across other benchmarks, confirming the adaptability and efficiency of LLM-designed KTMs in diverse EMTO configurations.

### IV. CONCLUSION

This paper introduces an LLM-assisted framework for autonomously designing KTMs in EMTO, mitigating the reliance on expert-crafted models. By combining few-shot prompting with multi-objective optimization, it balances transfer quality and efficiency. Experimental results on ten 50-task benchmarks demonstrate that the LLM-generated KTMs significantly outperform hand-crafted EMTO methods in both accuracy and runtime, highlighting the framework’s adaptability and scalability for EMTO.

In the future, we would like to extend this approach to broader EMTO landscapes and integrating more advanced prompt strategies to enhance model diversity and generalization in real-world applications.

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